

Conditional Filling-Probability Limit Order Execution for Futures Trading

An Order Management Rule for AI Agent Position Signals

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Abstract

This project studies the order execution problem for an AI futures trading agent using the given futures market data. The agent provides a target position every five minutes, while the futures data gives the market prices, volumes, and contract information needed to evaluate execution. The original benchmark assumes that every required position change is filled immediately at the observed price. We keep the AI agent’s signal, target position path, and trading direction fixed. The only part we change is the execution method used to fill each required trade.

Our final method is based on conditional filling probability. When the agent needs to buy, we place a buy limit order below the current price. When the agent needs to sell, we place a sell limit order above the current price. For each possible limit distance k , measured in ticks, we estimate the historical probability that the market would move far enough to fill the order. We then choose the largest k whose estimated filling probability is above a required threshold.

The final reported results are only based on conditional filling probability above. They are not based on a trend trading rule. Trend-type methods, such as volume and volatility regimes, were tested throughout the project, but they performed poorly on some assets, so they are not the source of the final result. Trend is used only as one possible state variable for conditioning the empirical filling distribution.

Across seven futures markets, the method improves PnL in EuroStoxx, Gold, Heating Oil, and Nasdaq. It is unchanged in German Bunds and underperforms in British Pound and Japanese Yen. The best percentage improvement is EuroStoxx, with a 68.27% gain over the agent benchmark. The best absolute improvement is Nasdaq, with a 4,633.75 PnL improvement. In aggregate, total strategy PnL is 39,183.31 compared with total agent PnL of 33,107.39, a total improvement of 6,075.93, or 18.35%. The main conclusion is that conditional passive execution can improve results in some markets, but it is not a universal rule.

1 Introduction

The AI agent already tells us what position it wants. At each five-minute timestamp, the agent has a target position. If the target position changes, the agent needs to trade. The original benchmark

assumes the trade happens immediately at the current observed price.

This project does not try to improve the AI agent’s prediction. We do not change whether the agent wants to buy or sell. We do not change the target position. We only study the order management problem:

Given the same AI agent position changes, can we execute the trades at better prices than immediate execution?

The idea is simple. If the agent needs to buy, we can try to buy at a slightly lower price. If the agent needs to sell, we can try to sell at a slightly higher price. This can improve the execution price. But if the order is placed too far away, the market may never reach that price and the order may not fill.

So the whole project is about choosing the limit order distance. A larger distance gives a better price if filled, but a lower probability of being filled. A smaller distance gives a higher probability of being filled, but less price improvement.

The final strategy uses conditional filling probability to choose this distance. It asks:

Given the current market state, how many ticks away can we place the order while still maintaining a target filling probability?

This is why the final method should be understood as an execution rule, not a new alpha signal. The agent still decides the position. Our strategy only decides the execution price.

It is also worth noting that we explored other methods for improving execution and price improvement, such as trend analysis, but the results and the final PnL were drastically better off with the conditional filling probability approach.

2 Data and Cleaning

2.1 Raw Data

The project uses two types of data.

First, we use futures OHLCV data. The raw futures data contains:

[time, open, high, low, close, volume].

Second, we use AI agent data. The agent data contains:

[date, hour, minute, price, position].

The agent position is observed every five minutes. The strategy uses the change in the agent’s target position (right inclusive) as the required trade size.

The final results uses seven futures markets, shown in Table 1.

Market	Asset Class
EuroStoxx	European equity index futures
GBP – British Pound	Currency futures
German Bunds	European rates futures
Gold	Precious metals futures
Heating Oil	Energy futures
JPY – Japanese Yen	Currency futures
Nasdaq	U.S. equity index futures

Table 1: Futures markets included in the final backtest.

2.2 Continuous Futures Series

Each futures market has multiple contracts with different expiration dates. A contract is most useful when it is liquid. Before it becomes the front contract, its volume is usually very low. Near expiration, volume often moves to the next contract.

The repository has two cleaning approaches. The first approach uses volume to identify the liquid period. The second approach uses the AI agent price to identify which futures contract the agent appears to be trading. We also briefly introduced a "coverage-ratio" approach, but it failed for 3 out of the 7 asset classes. The final trading analysis uses the second approach, called the agent rule in the repository. This is more consistent with the benchmark because the execution strategy and the AI agent should be compared on the same underlying contract.

After cleaning, the futures contracts are stitched into a continuous series. The final files are saved in:

`futures_right` and `agent_right`.

The one-minute futures data is used to calculate volatility features. The five-minute futures data is used for the actual trading loop and for matching the agent's five-minute position data.

2.3 Trading Dataset

The final trading dataset merges the cleaned five-minute futures data with the AI agent position data by timestamp. At each timestamp t , we observe:

$$(P_t, q_t),$$

where P_t is the current close price and q_t is the agent's target position.

The required trade size is:

$$\Delta q_t = q_t - q_{t-1}.$$

If $\Delta q_t > 0$, the agent needs to buy. If $\Delta q_t < 0$, the agent needs to sell. If $\Delta q_t = 0$, there is no new order.

3 Connection to the Project PDF

The project PDF defines range, range up, and range down over a future interval. For a horizon τ , the main quantities are:

$$R_{t,\tau} = H - L,$$

$$R_{t,\tau}^{(U)} = H - O,$$

$$R_{t,\tau}^{(D)} = O - L.$$

These quantities are directly related to limit order filling.

For a buy limit order, we care whether the price goes down after the order is placed. So the relevant quantity is the downward move. In our code, this is:

$$\text{future_low_move} = P_t - \min_{u \in (t, t+\tau]} P_u^{low}.$$

This is the execution version of range down.

For a sell limit order, we care whether the price goes up after the order is placed. So the relevant quantity is the upward move. In our code, this is:

$$\text{future_high_move} = \max_{u \in (t, t+\tau]} P_u^{high} - P_t.$$

This is the execution version of range up.

Therefore, the project does not need to forecast direction. For a buy order, it only estimates whether the future low is low enough to fill the buy limit order. For a sell order, it only estimates whether the future high is high enough to fill the sell limit order.

4 Benchmark PnL

The benchmark is the original AI agent execution. The benchmark assumes that every position change is executed immediately at the agent price, or at the corresponding close price when needed.

For each timestamp t , the benchmark cash flow is:

$$\text{cash flow}_t^{agent} = -\Delta q_t P_t.$$

Cumulative cash is:

$$\text{cash}_t^{\text{agent}} = \sum_{s \leq t} \text{cash flow}_s^{\text{agent}}.$$

The agent position is:

$$q_t = \sum_{s \leq t} \Delta q_s.$$

The marked-to-market benchmark PnL is:

$$\text{PnL}_t^{\text{agent}} = \text{cash}_t^{\text{agent}} + q_t P_t.$$

This benchmark is important because our strategy receives the exact same position changes. Any improvement comes only from execution price and timing.

5 Conditional Filling-Probability Rule

5.1 Candidate Limit Prices

At time t , suppose the current reference price is P_t and the tick size is δ .

If the agent needs to buy, candidate buy limit prices are:

$$P_{k,t}^{\text{buy}} = P_t - k\delta, \quad k = 0, 1, 2, \dots, K.$$

If the agent needs to sell, candidate sell limit prices are:

$$P_{k,t}^{\text{sell}} = P_t + k\delta, \quad k = 0, 1, 2, \dots, K.$$

The value k controls how passive the order is.

- Small k : closer to the current price, easier to fill, less price improvement.
- Large k : farther from the current price, harder to fill, more price improvement if filled.

In the final implementation, the maximum distance is:

$$K = 10.$$

5.2 Fill Probability for a Buy Order

For a buy order placed k ticks below the current price, the order fills if the future low reaches the target price. Equivalently:

$$\text{future_low_move}_{t,\tau} \geq k\delta.$$

So the estimated buy filling probability is:

$$\hat{p}_t^{buy}(k) = \widehat{\Pr}(\text{future_low_move}_{t,\tau} \geq k\delta \mid \mathcal{S}_t),$$

where $\mathcal{S}_t$ is the current market state.

5.3 Fill Probability for a Sell Order

For a sell order placed k ticks above the current price, the order fills if the future high reaches the target price. Equivalently:

$$\text{future_high_move}_{t,\tau} \geq k\delta.$$

So the estimated sell filling probability is:

$$\hat{p}_t^{sell}(k) = \widehat{\Pr}(\text{future_high_move}_{t,\tau} \geq k\delta \mid \mathcal{S}_t).$$

5.4 Choosing k

The strategy chooses the largest k whose estimated filling probability is at least the required threshold p^* .

For either a buy or sell order:

$$k^* = \max \{k \in \{0, 1, \dots, K\} : \hat{p}_t(k) \geq p^*\}.$$

Then the order is submitted at the corresponding target price.

This is the most important part of the project. The rule is not asking whether the agent is right. The rule is only asking how far away the limit order can be placed while still having a high enough historical filling probability.

6 State Variables

6.1 Why Use States?

Market conditions change over time. A high-volatility period should have different filling probabilities from a low-volatility period. If price usually moves more, then a deeper limit order may still have a reasonable chance of filling. If price barely moves, then only shallow orders are likely to fill.

For this reason, the strategy estimates conditional filling probabilities by market state.

6.2 Final State Variables

The final repository implementation uses two state variables:

1. **Volatility state**

2. Crossover state

The volatility feature is based on realized volatility from one-minute returns over the τ -minute horizon. It is then smoothed with a sparse EMA so that the feature uses non-overlapping τ -intervals.

The crossover feature is:

$$\text{crossover}_t = \text{EMA}_t^{\text{short}} - \text{EMA}_t^{\text{long}}.$$

The code also scales it by price:

$$\text{crossover_pct}_t = \frac{\text{crossover}_t}{P_t} \times 100.$$

The state space is built by dividing volatility into three quantile buckets and crossover into three quantile buckets. This gives:

$$3 \times 3 = 9$$

possible regimes.

A state might look like:

$$\text{vol_0_cross_2.}$$

This means low volatility and high crossover. The strategy then uses historical observations from the same state to estimate filling probabilities.

6.3 Important Clarification About Trend

The crossover variable is not used as a trend trading rule. It does not decide whether to buy or sell. The AI agent already decides the direction and position size.

The crossover variable only helps divide historical observations into different buckets. In other words, it is used to condition the filling distribution.

This distinction is important. The final results are only based on conditional filling probability. The trend method itself performed badly on some assets, so the final project should not be interpreted as evidence that a trend rule works.

6.4 Volume

Volume was studied as a possible state variable because the project PDF emphasizes volume and volatility. Volume is also useful for cleaning the futures data and identifying liquid contract periods.

However, the final trading rule does not use volume as a separate bucket. In the feature analysis, volatility already captured much of the useful information about future price range, and adding volume increased the number of states. This can make the conditional distributions too sparse. Therefore, the final implementation uses volatility and crossover states only.

This is one limitation of the current version. A future version could add volume back in a more stable way.

7 Fallback Rule

Some states do not have enough historical observations. If the current state has too few observations, the conditional estimate is not reliable.

The implementation uses:

$$\text{min_obs_per_regime} = 25.$$

If the current state has at least 25 observations, the strategy uses the state-specific distribution. If not, the strategy falls back to the global historical distribution.

So the rule is:

- Use the regime-conditional distribution when there is enough data.
- Use the global fallback distribution when the current regime is too sparse.

This makes the rule more stable.

8 Fixed Target Execution Rule

Once k is chosen, the strategy creates a fixed target price.

For a buy order:

$$P_{\text{target}} = P_t - k^* \delta.$$

For a sell order:

$$P_{\text{target}} = P_t + k^* \delta.$$

The word fixed means that the target price is chosen once and then kept fixed. The strategy does not reset the target price every five minutes. It does not recompute a new limit price after the first τ -minute window.

The execution logic is:

1. A new order is created when the agent position changes.
2. The strategy checks whether the target price is reached during the next τ minutes.
3. If the target is reached in that first window, the order is marked as successful.
4. If the target is not reached, the order stays active.

5. If the target is reached later, the order is marked as delayed execution.
6. If the target is never reached by the end of the backtest, the order is forced to execute at the final close.

For a buy order, the fill condition is:

$$L_s \leq P_{\text{target}}.$$

For a sell order, the fill condition is:

$$H_s \geq P_{\text{target}}.$$

This fixed rule produced the final results. It can improve PnL because it waits for better prices, but it also creates stale-order risk.

9 Look-Ahead Bias Control

The strategy must avoid using information from the future.

For any historical timestamp u , the future τ -minute move is not known until $u + \tau$. Therefore, a sample from time u can only be added to the empirical distribution after:

$$u + \tau \leq t.$$

In the implementation, the distribution is initialized with preparation data. During the trading period, the distribution is updated once per day. At the start of a new trading day, the strategy adds only the outcomes that were already observable before the previous day ended.

This prevents look-ahead bias. Without this rule, the strategy would use future high-low information that would not have been known at the decision time.

10 Full Workflow

The full workflow is shown in Figure 1.

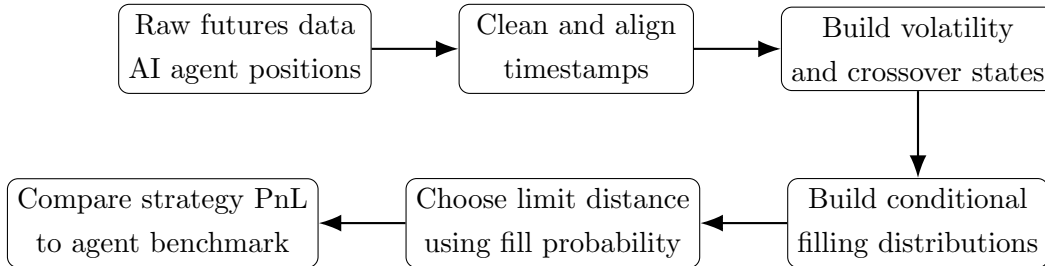


Figure 1: Final execution workflow. The final rule is based on conditional filling probability, not a trend trading signal.

11 Experimental Design

The experiment compares two execution methods:

1. **Agent benchmark:** execute every agent position change immediately.
2. **Conditional filling-probability strategy:** place a limit order at the deepest tick distance whose estimated filling probability is above the required threshold.

Each market is tested separately. The final parameters are shown in Table 2.

Market	τ Window	Fill-Probability Threshold
EuroStoxx	60 minutes	0.50
GBP – British Pound	5 minutes	0.90
German Bunds	5 minutes	0.90
Gold	20 minutes	0.80
Heating Oil	60 minutes	0.50
JPY – Japanese Yen	5 minutes	0.90
Nasdaq	10 minutes	0.90

Table 2: Market-specific execution parameters used in the final backtest.

The main evaluation metric is:

$$\text{PnL Difference} = \text{Strategy PnL} - \text{Agent PnL}.$$

The percentage improvement is:

$$\text{Improvement} = \frac{\text{Strategy PnL} - \text{Agent PnL}}{\text{Agent PnL}}.$$

Since the agent’s position path is unchanged, this difference measures execution value added.

12 Results

12.1 Final PnL Comparison

Table 3 reports the final PnL comparison.

Market	Agent PnL	Strategy PnL	Difference	Improvement
EuroStoxx	1,219.50	2,052.00	832.50	68.27%
GBP – British Pound	92.36	73.12	-19.24	-20.83%
German Bunds	-10.48	-10.48	0.00	0.00%
Gold	1,882.00	2,163.30	281.30	14.95%
Heating Oil	966.82	1,314.51	347.69	35.96%
JPY – Japanese Yen	6.94	6.86	-0.07	-1.08%
Nasdaq	28,950.25	33,584.00	4,633.75	16.01%
Total	33,107.39	39,183.31	6,075.93	18.35%

Table 3: Final PnL comparison between immediate agent execution and conditional filling-probability execution.

The strategy improves PnL in four markets:

- EuroStoxx
- Gold
- Heating Oil
- Nasdaq

It is unchanged in German Bunds and underperforms in British Pound and Japanese Yen.

EuroStoxx has the best percentage improvement. The strategy improves PnL by 832.50, or 68.27%.

Nasdaq has the best absolute improvement. The strategy improves PnL by 4,633.75, or 16.01%.

Across all seven markets, total agent PnL is 33,107.39 and total strategy PnL is 39,183.31. This gives a total PnL improvement of 6,075.93, or 18.35%.

These results show that the method can help, but only in some markets. The rule works when the market usually moves enough for passive limit orders to fill. It does not help much when the conditional filling distribution forces $k = 0$ most of the time.

12.2 Execution Detail

Table 4 shows the execution details.

Market	Trades	Successful	Delayed	Forced Close	Mean k
EuroStoxx	212	169	43	0	10.00
GBP – British Pound	449	422	17	10	0.00
German Bunds	613	595	18	0	0.00
Gold	1,276	1,138	134	4	1.00
Heating Oil	290	167	88	35	3.90
JPY – Japanese Yen	762	734	22	6	0.00
Nasdaq	2,164	2,042	122	0	2.42

Table 4: Execution-state summary from the final backtest.

The execution table helps explain the PnL results.

Nasdaq has many trades, a positive average k , and no forced closes. This is why it produces the largest absolute improvement.

Gold has an average k of about one tick and only four forced closes. This is a clean example where small price improvement across many trades improves total PnL.

Heating Oil has a larger average k , but also many forced closes. This means the strategy is more aggressive. It captures better prices when it fills, but it also has more stale-order risk.

British Pound, German Bunds, and Japanese Yen all have mean $k = 0$. This means the conditional filling rule usually decides that deeper passive orders do not satisfy the required filling threshold. In those markets, there is little room for execution improvement.

12.3 Market-by-Market Interpretation

The results are market-specific.

EuroStoxx improves the most in percentage terms. But it uses a 60-minute window and a 0.50 filling threshold. This is a loose setting. The strategy is allowed to wait longer and accept a lower estimated fill probability.

Nasdaq is the strongest absolute result. It has many trades, positive average order depth, and no forced closes. This makes the execution improvement more convincing.

Gold is more moderate. The strategy usually chooses only about one tick of price improvement, but it does this consistently and with few forced closes.

Heating Oil improves PnL, but the execution quality is less clean because there are 35 forced closes. This market shows the tradeoff of the method: deeper orders can make more money when they fill, but they also create more risk when they do not fill.

British Pound and Japanese Yen do not improve. German Bunds is unchanged. These markets show why the final result cannot be described as a universal rule. The method depends on the empirical filling distribution of each market.

13 Risks and Limitations

13.1 Stale Orders

The biggest risk is stale orders. If an order does not fill during the first τ -minute window, the current strategy leaves it active. This can improve the backtest if the market later returns to the target price. But in live trading, the agent's desired position may change before the old order fills.

This means a stale order can move the strategy away from the current target position.

A more realistic version should cancel orders more quickly. Possible cancellation rules include:

- cancel after one τ -minute window,
- cancel after a fixed number of bars,
- cancel when the agent changes direction,
- cancel when the estimated filling probability changes too much.

13.2 Fill Assumption

The backtest assumes that a limit order fills if the bar high or low reaches the target price.

For a buy order, the fill condition is:

$$L_s \leq P_{\text{target}}.$$

For a sell order, the fill condition is:

$$H_s \geq P_{\text{target}}.$$

This is a useful first-pass assumption, but it is optimistic. In real markets, touching a price does not guarantee a fill. Actual execution depends on queue position, order priority, available liquidity, and traded volume at the price.

So the results should be interpreted as a bar-data execution simulation, not as a complete live-trading estimate.

13.3 No Partial Fills

The strategy assumes the full order fills once the target price is reached. In real futures markets, an order may be partially filled. A more realistic simulator should allow part of the order to fill while the remaining quantity stays open.

13.4 Simplified Tick Size

The final script uses a simplified tick-size treatment. In reality, each futures contract has its own tick size, point value, and dollar tick value. A one-tick improvement in Nasdaq is not the same economic amount as a one-tick improvement in Gold or Heating Oil.

A better version should use contract-specific tick sizes, tick values, multipliers, and fee schedules.

13.5 Transaction Costs and Market Impact

The current backtest does not fully model bid-ask spread, commissions, exchange fees, market impact, or queue position. These costs would reduce realized PnL.

Passive execution can improve price, but missed orders and delayed orders also create opportunity cost. This is why the current results should be treated as evidence of potential execution improvement, not as final production trading results.

13.6 Not a Universal Strategy

The method does not work equally well across all assets. It improves Nasdaq, EuroStoxx, Gold, and Heating Oil, but it does not improve British Pound, Japanese Yen, or German Bunds.

This is the main lesson of the project. Conditional filling probability can improve execution when the market has enough short-term range. But if the market does not support passive orders at the required filling threshold, the rule often chooses $k = 0$ and becomes close to immediate execution.

14 Conclusion

This project studies order management for an AI futures trading agent. The agent's target positions are fixed. We do not change the signal. We only change how the required position changes are executed.

The final method is a conditional filling-probability rule. For each buy or sell order, the strategy estimates how likely the market is to move favorably by at least k ticks. It then chooses the largest k that still satisfies the required filling probability threshold.

The final results are not based on a trend method. Trend-style methods were tested during development, but they performed poorly on some assets. In the final implementation, the crossover variable is only used as a conditioning variable for the empirical filling distribution. It does not determine trading direction.

The results show that the method can improve execution in some markets. EuroStoxx improves by 68.27%, Heating Oil by 35.96%, Gold by 14.95%, and Nasdaq by 16.01%. Across all seven markets, total PnL improves by 6,075.93, or 18.35%. Nasdaq gives the largest absolute PnL improvement of 4,633.75.

The weaker markets are also important. British Pound underperforms, Japanese Yen is slightly negative, and German Bunds is unchanged. These results show that the rule is not universal. It works when the historical range behavior of the market supports passive limit orders.

Future work should focus on stricter cancellation rules, contract-specific tick values, transaction costs, partial fills, queue position, and order-book-level fill modeling. These changes would make

the backtest more realistic and would show whether the same conditional filling-probability idea can work in a production trading system.